Data Challenge Scenarios: Solving Real-World Machine Learning Problems

Introduction

This artifact summarizes my experience completing the Data Challenge Scenarios activity using the SchoolAI coach. The activity presented three real-world machine learning scenarios that required me to analyze data challenges, explain my reasoning, and improve my decisions through continuous feedback. Instead of simply answering questions, I evaluated different approaches and learned how data-related decisions can affect machine learning model performance.

Scenario 1: Handling Missing Data

The first scenario involved a healthcare dataset where approximately 15% of laboratory results were missing. At first, I considered removing records with missing values or collecting additional data. Through the discussion, I learned that removing too many records could introduce bias and that historical data cannot simply be recollected. I explored data imputation using similar patient information and learned that the median is often a better choice than the mean for medical data because it is less affected by extreme values.

Key Takeaways

- Removing missing data can introduce bias.
- Data imputation helps preserve more training data.
- Data cleaning requires balancing data quality with model accuracy.

Scenario 2: Handling Imbalanced Data

The second scenario focused on a fraud detection system where only 0.8% of transactions were fraudulent. I learned that a model can achieve high accuracy while still failing to detect fraud because of the imbalance between legitimate and fraudulent transactions. I explored undersampling, oversampling, and synthetic data generation, and learned that precision, recall, and F1-score are more appropriate evaluation metrics than accuracy for this type of problem.

Key Takeaways

- Accuracy alone is not enough for imbalanced datasets.
- Sampling techniques can improve minority class detection.
- Evaluation metrics should match the business objective.

Scenario 3: Handling Data Drift

The final scenario involved a credit scoring model whose performance declined after economic conditions changed. I learned that changes in customer behavior and market conditions can reduce model accuracy over time. I explored detecting data drift, retraining models with recent data, and validating improvements before deployment.

Key Takeaways

- Machine learning models require continuous monitoring.
- Data drift can significantly reduce prediction accuracy.
- Retraining should balance historical data with recent trends.

Summary

This activity helped me understand that successful machine learning depends not only on selecting the right algorithm but also on making informed decisions about the data. Working through these scenarios strengthened my problem-solving and critical-thinking skills while improving my understanding of missing data, imbalanced datasets, and data drift in real-world machine learning systems.